

Nash Bargaining SQP+OptNet Layer: Mathematical Formulation

1 Decision Variables

For a family with M members, each member m has at most T activities. The per-activity decision vector is:

$$\mathbf{x}_{m,t} = [s_{m,t}, e_{m,t}, p_{m,t,1}^{\text{mode}}, \dots, p_{m,t,K}^{\text{mode}}, j_{m,t}, d_{m,t}] \in \mathbb{R}^V, \quad (1)$$

where $V = 2 + K + 2 = 15$ (with $K = 11$ travel modes), and:

- $s_{m,t}, e_{m,t}$: start and end time in z-score normalization ($\mu = 12.946$ h, $\sigma = 4.691$ h),
- $p_{m,t,k}^{\text{mode}}$: probability of travel mode k , forming a simplex $\sum_{k=1}^K p_{m,t,k}^{\text{mode}} = 1$,
- $j_{m,t}$: joint travel probability ($\in [\varepsilon, 1 - \varepsilon]$),
- $d_{m,t}$: driver probability ($\in [\varepsilon, 1 - \varepsilon]$), where $\varepsilon = 10^{-3}$.

The full family decision vector is obtained by concatenating all per-member, per-activity vectors:

$$\mathbf{x} = [\mathbf{x}_{1,1}, \dots, \mathbf{x}_{1,T}, \dots, \mathbf{x}_{M,1}, \dots, \mathbf{x}_{M,T}] \in \mathbb{R}^n, \quad n = M \times T \times V. \quad (2)$$

2 Utility Function

The utility of member m is defined as:

$$\begin{aligned} U_m(\mathbf{x}) = & \beta_0 + \alpha_{\text{social}} \sum_{t=1}^T \tilde{o}_{m,t} a_{m,t} + \alpha_{\text{joint}} \sum_{t=1}^T j_{m,t} a_{m,t} + \theta_{\text{joint_cost}} \sum_{t=1}^T j_{m,t}^2 a_{m,t} \\ & + \theta_{\text{escort}} \sum_{t=1}^T d_{m,t} (e_{m,t} - s_{m,t}) a_{m,t} \mathbb{K}_{\text{adult}}(m) \\ & + \sum_{t=1}^T \tilde{o}_{m,t} \left(\sum_{k=1}^K p_{m,t,k}^{\text{mode}} \theta_{\text{travel},k} \mathbb{E}[\text{TT}_{m,t,k}] \right) a_{m,t}, \end{aligned} \quad (3)$$

where:

- β_0 : learnable baseline utility,
- α_{social} : learnable social (out-of-home) preference weight,
- α_{joint} : learnable joint travel preference weight,
- $\theta_{\text{joint_cost}}$: learnable quadratic joint coordination cost (typically negative),
- θ_{escort} : learnable escort burden coefficient (typically negative),
- $\theta_{\text{travel},k}$: learnable mode-specific travel time disutility (vector of length K),

- $\tilde{o}_{m,t} = 1 - P(\text{dest}_{m,t} = \text{home})$: out-of-home probability from the destination head (fixed, not optimized),
- $\mathbb{E}[\text{TT}_{m,t,k}]$: expected travel time for mode k at activity t , derived from the travel time matrix and destination probabilities (fixed),
- $a_{m,t}$: activity validity mask (1 if activity t of member m is valid, 0 otherwise),
- $\mathbb{K}_{\text{adult}}(m)$: indicator for adult members.

Quadratic joint coordination cost. The quadratic term $\theta_{\text{joint_cost}} \cdot j_{m,t}^2$ prevents corner solutions in the joint travel probability. With $\theta_{\text{joint_cost}} < 0$ (a cost) and $\alpha_{\text{joint}} > 0$ (a benefit), the first-order condition for the optimal joint probability is:

$$j_{m,t}^* = \frac{\alpha_{\text{joint}}}{2|\theta_{\text{joint_cost}}|}, \quad (4)$$

yielding an interior optimum rather than a boundary solution at 0 or 1. For example, with $\alpha_{\text{joint}} = 0.1$ and $\theta_{\text{joint_cost}} = -0.5$, the unconstrained optimum is $j^* \approx 0.1$.

Lower bound. To ensure positivity for the log-utility in the Nash welfare function, a smooth lower bound is applied:

$$U_m \leftarrow U_m + \text{softplus}(1 - U_m), \quad (5)$$

where $\text{softplus}(z) = \log(1 + e^z)$. This guarantees $U_m \geq 1$ smoothly without gradient discontinuity. Utilities of invalid members are set to β_0 to avoid affecting the Nash objective.

3 V5: Context-Aware Utility Parameters

In the base model, the utility parameters α_{joint} , α_{social} , and θ_{escort} are global scalars shared across all family members and activities. V5 introduces a **UtilityParameterNet** that predicts per-activity parameter values conditioned on the family context:

$$[\alpha_{\text{joint}}^{\text{ctx}}, \alpha_{\text{social}}^{\text{ctx}}, \theta_{\text{escort}}^{\text{ctx}}]_{m,t} = f_{\text{MLP}}(\mathbf{f}_{\text{family}}, \mathbf{f}_{\text{member},m}, \mathbf{p}_{m,t}^{\text{purpose}}), \quad (6)$$

where $\mathbf{f}_{\text{family}} \in \mathbb{R}^{F_f}$ and $\mathbf{f}_{\text{member},m} \in \mathbb{R}^{F_m}$ are the family and member attribute vectors, and $\mathbf{p}_{m,t}^{\text{purpose}} \in \mathbb{R}^{10}$ is the predicted activity purpose probability (detached from the gradient graph).

Network architecture. The MLP consists of two linear layers with a single hidden layer of dimension 64:

$$\mathbb{R}^{F_f+F_m+10} \xrightarrow{\text{Linear}(64)} \xrightarrow{\text{ReLU}} \xrightarrow{\text{Dropout}(0.1)} \xrightarrow{\text{Linear}(3)} \mathbb{R}^3. \quad (7)$$

The three outputs are transformed via activation functions that enforce the correct sign:

$$\alpha_{\text{joint}}^{\text{ctx}} = \text{softplus}(z_0) \times 0.5, \quad (8)$$

$$\alpha_{\text{social}}^{\text{ctx}} = \text{softplus}(z_1) \times 0.5, \quad (9)$$

$$\theta_{\text{escort}}^{\text{ctx}} = -\text{softplus}(z_2), \quad (10)$$

where $[z_0, z_1, z_2]$ are the raw MLP outputs. The softplus activation ensures $\alpha_{\text{joint}}^{\text{ctx}}, \alpha_{\text{social}}^{\text{ctx}} > 0$ (preferences) and $\theta_{\text{escort}}^{\text{ctx}} < 0$ (burden).

Initialization. The MLP weights are initialized with small magnitude ($\times 0.01$), and the output biases are set so that the initial predictions approximate the global parameter values. Specifically, for a target value α^* with activation $\text{softplus}(b) \times 0.5 \approx \alpha^*$, the bias is initialized as $b = \log(\exp(2\alpha^*) - 1)$. This ensures the model starts near the baseline behavior and smoothly deviates as training progresses.

Integration with utility function. When context-aware parameters are enabled, the scalar parameters in Eq. (3) are replaced by their per-activity counterparts: $\alpha_{\text{social}} \rightarrow \alpha_{\text{social},m,t}^{\text{ctx}}$, $\alpha_{\text{joint}} \rightarrow \alpha_{\text{joint},m,t}^{\text{ctx}}$, $\theta_{\text{escort}} \rightarrow \theta_{\text{escort},m,t}^{\text{ctx}}$. The global scalar parameters remain in the model and serve as the initial reference point.

Gradient flow. During Phase 1 (SQP), the context-aware parameters are detached from the computation graph to avoid accumulating gradients through the non-differentiable SQP iterations. Gradients flow through the UtilityParameterNet only in Phase 2 (OptNet), via the Q_{diag} term in the diagonal KKT backward (Eq. 47):

$$\frac{\partial \mathcal{L}}{\partial \phi_{\text{MLP}}} = \frac{\partial \mathcal{L}}{\partial \mathbf{d}^*} \cdot \frac{\partial \mathbf{d}^*}{\partial Q_{\text{diag}}} \cdot \frac{\partial Q_{\text{diag}}}{\partial U_m^2} \cdot \frac{\partial U_m}{\partial \alpha^{\text{ctx}}} \cdot \frac{\partial \alpha^{\text{ctx}}}{\partial \phi_{\text{MLP}}}, \quad (11)$$

where ϕ_{MLP} denotes the MLP parameters.

4 V5: Dynamic Disagreement Point

In the base model, the disagreement point (fallback utility when no agreement is reached) is a learnable scalar β_0 shared by all members. V5 replaces this with a **dynamic, per-member disagreement point** that reflects each member’s utility under the decoder’s original (uncoordinated) prediction:

$$U_m^{\text{base}} = U_m(\boldsymbol{\theta}), \quad \beta_{0,m} = \max(U_m^{\text{base}}, 1.0), \quad (12)$$

where $\boldsymbol{\theta}$ is the decoder output packed into decision-variable format, and $U_m(\cdot)$ is the utility function (Eq. 3). The $\max(\cdot, 1.0)$ clamp ensures the baseline utility is strictly positive, which is necessary for the log-utility in the Nash welfare function.

Economic interpretation. In Nash Bargaining theory, the disagreement point represents the outside option—the utility each player receives if negotiation fails. Setting it to $U_m(\boldsymbol{\theta})$ means the “no coordination” outcome corresponds to the decoder’s independent prediction for each member. The Nash Bargaining solution then only adjusts activities when doing so improves *all* members above their uncoordinated baselines.

Integration. When dynamic baseline is enabled, the shared scalar β_0 in Eq. (3) is replaced by the per-member value $\beta_{0,m}$, so the utility becomes:

$$U_m(\mathbf{x}) = \beta_{0,m} + \alpha_{\text{social}} \sum_t (\cdots) + \cdots \quad (13)$$

The Nash welfare objective retains the same form $W(\mathbf{x}) = \sum_{m=1}^M \log U_m(\mathbf{x})$, but the member-specific baseline $\beta_{0,m}$ changes the utility landscape: members with higher uncoordinated utility (larger $\beta_{0,m}$) have a higher floor, giving them more bargaining power in the Nash solution. The dynamic baseline is computed once (at the decoder output) and detached; it does not receive gradients.

5 Nash Welfare Objective

The Nash Bargaining solution maximizes the product of member utilities, equivalently the sum of log-utilities:

$$\max_{\mathbf{x}} W(\mathbf{x}) = \sum_{m=1}^M \log U_m(\mathbf{x}). \quad (14)$$

The objective $W(\mathbf{x})$ is **nonlinear**: the $\log(\cdot)$ composition is inherently nonlinear, and the escort term in U_m contains the bilinear product $d_{m,t}(e_{m,t} - s_{m,t})$.

6 Phase 1: Nonlinear Program

Phase 1 solves the full nonlinear program (NLP) to find a converged point $\mathbf{x}_{\text{conv}}$.

6.1 Optimization Problem

$$\max_{\mathbf{x}} W(\mathbf{x}) = \sum_{m=1}^M \log U_m(\mathbf{x}) \quad (15)$$

subject to the constraints in Table 1.

Table 1: Phase 1 constraints in variable space $\mathbf{x}$. Constraints C1–C3 are enforced in all modes (via projection and, when enabled, in the SQP subproblem). Constraints C4–C5 are additionally included when V5 constrained SQP is enabled.

Constraint	Formulation	Type	Mode
(C1) Variable bounds	$x_i^{\text{lb}} \leq x_i \leq x_i^{\text{ub}}$	Linear (ineq)	All
(C2) Time ordering	$e_{m,t} \geq s_{m,t}, \quad s_{m,t+1} \geq e_{m,t}$	Linear (ineq)	All
(C3) Mode simplex	$\sum_{k=1}^K p_{m,t,k}^{\text{mode}} = 1$	Linear (eq)	All
(C4) Vehicle capacity	$\sum_{m,t} d_{m,t} \cdot p_{m,t}^{\text{car}} \cdot \mathbb{1}_{\text{adult}} \cdot a_{m,t} \leq n_{\text{veh}}$	Bilinear $\rightarrow$ linearized (ineq)	V5
(C5) Joint consistency	Gate-activated pairwise attribute matching	Nonlinear $\rightarrow$ gate-fixed (eq)	V5

The variable bounds (C1) are:

$$s_{m,t}, e_{m,t} \in [z_{\min}, z_{\max}], \quad z_{\min} = \frac{1 - \mu}{\sigma}, \quad z_{\max} = \frac{24 - \mu}{\sigma}, \quad (16)$$

$$p_{m,t,k}^{\text{mode}} \in [\varepsilon, 1 - \varepsilon], \quad (17)$$

$$j_{m,t}, d_{m,t} \in [\varepsilon, 1 - \varepsilon]. \quad (18)$$

Scope of Phase 1 constraints. In the original mode, Phase 1 handles only the per-variable and per-activity constraints (C1–C3) via projection after each SQP step. When V5 constrained SQP is enabled (USE_CONSTRAINED_SQP), all five constraints (C1–C5) are enforced within the QP subproblem at each SQP iteration (see §6.2). In this mode, the vehicle capacity constraint (C4) is linearized at the current iterate $\mathbf{x}^{(i)}$, and the joint consistency constraint (C5) uses a sigmoid gate evaluated and fixed at $\mathbf{x}^{(i)}$. In both modes, the projection operator additionally enforces vehicle and joint consistency heuristically (see Projection below). Phase 2 (§7) re-expresses all five constraints in the displacement space $\mathbf{d} = \mathbf{x} - \mathbf{x}_{\text{conv}}$, linearized at the converged point $\mathbf{x}_{\text{conv}}$.

6.2 SQP Algorithm

Phase 1 solves the NLP using Sequential Quadratic Programming (SQP) with BFGS inverse Hessian approximation. In the original mode, constraints (C1–C3) are handled via projection after each step. In V5 constrained SQP mode, all five constraints (C1–C5) are enforced both in the QP subproblem and via projection.

V5: Constrained SQP subproblem. When USE_CONSTRAINED_SQP is enabled, the search direction at each SQP iteration is obtained by solving a constrained QP subproblem:

$$\min_{\mathbf{d}} \frac{1}{2} \mathbf{d}^\top (H + \delta I) \mathbf{d} + \mathbf{g}^\top \mathbf{d} \quad \text{s.t.} \quad G\mathbf{d} \leq \mathbf{h}' - G\mathbf{x}, \quad A\mathbf{d} = \mathbf{b}' - A\mathbf{x}, \quad (19)$$

Algorithm 1 Phase 1: SQP with BFGS and Armijo Line Search

Require: Decoder output θ , masks, utility parameters, context-aware parameters (V5, detached)

Ensure: Converged point $\mathbf{x}_{\text{conv}}$

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1:  $\mathbf{x} \leftarrow \theta$  ▷ Initialize at decoder output
2:  $H \leftarrow I$  ▷ Initial inverse Hessian approximation
3: for  $i = 0, \dots, I_{\max} - 1$  do
4:    $\mathbf{g} \leftarrow -\nabla_{\mathbf{x}} W(\mathbf{x})$  ▷ Negative gradient of Nash welfare
5:   if  $\mathbf{g}$  contains NaN then reset  $\mathbf{x} \leftarrow \theta$ ; break
6:   end if
7:   if  $\|\mathbf{g}\| < \epsilon_{\text{tol}}$  then break ▷ Converged
8:   end if
9:   if  $i > 0$  then
10:     $\mathbf{s} \leftarrow \mathbf{x}^{(i)} - \mathbf{x}^{(i-1)}, \quad \mathbf{y} \leftarrow \mathbf{g}^{(i)} - \mathbf{g}^{(i-1)}$ 
11:     $H \leftarrow \text{BFGS-UPDATE}(H, \mathbf{s}, \mathbf{y})$  ▷ With Powell's damping
12:   end if
13:   if USE_CONSTRAINED_SQP then ▷ — Search direction (V5 branching) —
14:      $\mathbf{p} \leftarrow \arg \min_{\mathbf{d}} \frac{1}{2} \mathbf{d}^\top (H + \delta I) \mathbf{d} + \mathbf{g}^\top \mathbf{d} \text{ s.t. } G\mathbf{d} \leq \mathbf{h}', A\mathbf{d} = \mathbf{b}'$  ▷ V5: Constrained QP subproblem
15:     if QP fails or returns NaN then ▷ Eq. (19)
16:        $\mathbf{p} \leftarrow -(H + \delta I) \mathbf{g}$  ▷ Fallback
17:     end if
18:   else ▷ Original: unconstrained Newton step
19:      $\mathbf{p} \leftarrow -(H + \delta I) \mathbf{g}$  ▷  $\delta = 10^{-3}$ 
20:   end if
21:    $\alpha \leftarrow \text{ARMIJO-SEARCH}(\mathbf{x}, \mathbf{p})$  ▷ Backtracking line search
22:    $\mathbf{x} \leftarrow \text{PROJECT}(\mathbf{x} + \alpha \mathbf{p})$  ▷ Project onto (C1)–(C3) + vehicle + joint
23:   if  $\mathbf{x}$  contains NaN then reset  $\mathbf{x} \leftarrow \theta$ ; break
24:   end if
25: end for
26: return  $\mathbf{x}_{\text{conv}} \leftarrow \mathbf{x}$ 
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where $G, \mathbf{h}', A, \mathbf{b}'$ encode all five constraints (C1–C5) expressed as linear constraints on the step $\mathbf{d}$ at the current iterate $\mathbf{x}$: C1 (bounds), C2 (time ordering), C3 (mode simplex), C4 (vehicle capacity, linearized), and C5 (joint consistency, gate-fixed). The detailed formulations of C4 and C5 in the SQP subproblem are given below (§6.2). The inequality bounds are relaxed by 10^{-4} for numerical feasibility. QR rank filtering (as in Algorithm 2) is applied to the equality constraints before solving. If the QP solver fails or returns NaN/Inf values, the algorithm falls back to the unconstrained Newton direction $\mathbf{p} = -(H + \delta I)\mathbf{g}$ and relies on projection to maintain feasibility.

SQP subproblem constraints C4–C5. The SQP subproblem includes all five constraints. Constraints C1–C3 on the step $\mathbf{d}$ take the same form as in Phase 2 (Eqs. 33–37), with $\mathbf{x}_{\text{conv}}$ replaced by the current iterate $\mathbf{x}^{(i)}$. Constraints C4 and C5 are formulated as follows.

(C4) Vehicle capacity (linearized at $\mathbf{x}^{(i)}$). The vehicle constraint involves the bilinear product $d_{m,t} \cdot p_{m,t}^{\text{car}}$, where both driver probability and car mode probability are decision variables. At the current iterate $\mathbf{x}^{(i)}$, the constraint is linearized via first-order Taylor expansion:

$$\sum_{m,t} \left(p_{m,t}^{\text{car},(i)} d_{m,t} + d_{m,t}^{(i)} p_{m,t}^{\text{car}} \right) \mathbb{1}_{\text{adult}}(m) a_{m,t} \leq n_{\text{veh}} - \sum_{m,t} d_{m,t}^{(i)} p_{m,t}^{\text{car},(i)} \mathbb{1}_{\text{adult}}(m) a_{m,t}, \quad (20)$$

where superscript (i) denotes values at the current SQP iterate, $d_{d_{m,t}}$ and $d_{p_{m,t}^{\text{car}}}$ are the step components for driver probability and car mode probability respectively, and $a_{m,t}$ is the activity validity mask. This contributes a single inequality row per sample. The key difference from the Phase 2 form (Eq. 36) is that linearization occurs at the *current* iterate $\mathbf{x}^{(i)}$ (which changes each SQP step) rather than the fixed converged point $\mathbf{x}_{\text{conv}}$.

(C5) Joint travel consistency (gate-fixed at $\mathbf{x}^{(i)}$). For each member pair (m_1, m_2) and activity t , a sigmoid gate determines whether joint consistency is enforced:

$$w_{m_1, m_2, t}^{(i)} = \mathbb{I}[\sigma(10(j_{m_1, t}^{(i)} \cdot j_{m_2, t}^{(i)} - \tau)) > 0.1], \quad (21)$$

where $\tau = 0.5$ and superscript (i) denotes evaluation at the current iterate. For each active pair ($w_{m_1, m_2, t}^{(i)} = 1$), the step direction must drive the paired variables toward agreement:

$$d_{s_{m_1, t}} - d_{s_{m_2, t}} = s_{m_2, t}^{(i)} - s_{m_1, t}^{(i)}, \quad (22)$$

$$d_{e_{m_1, t}} - d_{e_{m_2, t}} = e_{m_2, t}^{(i)} - e_{m_1, t}^{(i)}, \quad (23)$$

$$d_{p_{m_1, t, k}^{\text{mode}}} - d_{p_{m_2, t, k}^{\text{mode}}} = p_{m_2, t, k}^{\text{mode}, (i)} - p_{m_1, t, k}^{\text{mode}, (i)}, \quad \forall k, \quad (24)$$

$$d_{j_{m_1, t}} - d_{j_{m_2, t}} = j_{m_2, t}^{(i)} - j_{m_1, t}^{(i)}. \quad (25)$$

Each active pair contributes $2 + K + 1 = 14$ equality rows. The right-hand sides ensure that $\mathbf{x}^{(i)} + \mathbf{d}$ yields equal attribute values for the paired members. Unlike Phase 2 (Eqs. 39–42), the gate and linearization point are re-evaluated at each SQP iteration, allowing the active set of constrained pairs to evolve as the solution progresses.

BFGS inverse Hessian update with Powell’s damping. The matrix H approximates $[\nabla^2(-W)]^{-1}$, updated via the damped BFGS formula. Given step $\mathbf{s} = \mathbf{x}^{(i)} - \mathbf{x}^{(i-1)}$ and gradient change $\mathbf{y} = \mathbf{g}^{(i)} - \mathbf{g}^{(i-1)}$:

1. If $\mathbf{s}^\top \mathbf{y} < 0.2 \mathbf{s}^\top H \mathbf{s}$, apply Powell’s damping:

$$\phi = \frac{0.8 \mathbf{s}^\top H \mathbf{s}}{\mathbf{s}^\top H \mathbf{s} - \mathbf{s}^\top \mathbf{y}}, \quad \mathbf{y}_d = \phi \mathbf{y} + (1 - \phi) H \mathbf{s}. \quad (26)$$

Otherwise $\mathbf{y}_d = \mathbf{y}$.

2. Apply the BFGS update:

$$H^+ = \left(I - \rho \mathbf{s} \mathbf{y}_d^\top \right) H \left(I - \rho \mathbf{y}_d \mathbf{s}^\top \right) + \rho \mathbf{s} \mathbf{s}^\top, \quad \rho = \frac{1}{\mathbf{s}^\top \mathbf{y}_d}. \quad (27)$$

Armijo line search. Starting from $\alpha = 1$, backtrack with factor 0.5 until the sufficient decrease condition holds:

$$-W(\text{PROJECT}(\mathbf{x} + \alpha \mathbf{p})) < -W(\mathbf{x}) - c \alpha \|\mathbf{p}\|^2, \quad (28)$$

where $c = 10^{-4}$ is the Armijo parameter.

Projection. The projection operator $\text{PROJECT}(\cdot)$ enforces constraints (C1)–(C3) and heuristic versions of (C4)–(C5):

1. **(C1)** Clamp times: $s_{m,t}, e_{m,t} \in [z_{\min}, z_{\max}]$.
2. **(C2)** Enforce temporal ordering: for $t = 1, \dots, T$ sequentially, set $s_{m,t} \geq e_{m,t-1}$ and $e_{m,t} \geq s_{m,t}$.

3. **(C3)** Project mode probabilities onto the simplex: clamp to $[\varepsilon, \infty)$ and renormalize.
4. Clamp $j_{m,t}, d_{m,t} \in [\varepsilon, 1 - \varepsilon]$.
5. **(C4) Vehicle projection:** if $\sum_{m,t} d_{m,t} \cdot p_{m,t}^{\text{car}} \cdot \mathbb{K}_{\text{adult}}(m) \cdot a_{m,t} > n_{\text{veh}}$, scale down $d_{m,t}$ proportionally so that the total driver usage equals n_{veh} .
6. **(C5) Joint consistency projection** (2-member families): for each active joint pair (determined by the same sigmoid gate as Eq. 21), average the start time, end time, and mode probabilities of the two members. For families with 3 or more members, joint consistency is handled exclusively by the QP equality constraints.

7 Phase 2: Quadratic Program

Phase 2 constructs a QP around the converged point $\mathbf{x}_{\text{conv}}$ from Phase 1. The displacement variable is $\mathbf{d} = \mathbf{x} - \mathbf{x}_{\text{conv}}$. The QP enforces the same five constraints (C1–C5) as V5 constrained SQP, but expressed in the displacement space $\mathbf{d}$ and linearized at the *fixed* converged point $\mathbf{x}_{\text{conv}}$ (rather than the changing iterate $\mathbf{x}^{(i)}$).

7.1 QP Formulation

$$\min_{\mathbf{d}} \frac{1}{2} \mathbf{d}^\top Q \mathbf{d} + \mathbf{p}^\top \mathbf{d} \quad \text{s.t.} \quad G \mathbf{d} \leq \mathbf{h}, \quad A \mathbf{d} = \mathbf{b}. \quad (29)$$

Quadratic term Q . Q is a diagonal matrix encoding the Nash Hessian approximation plus anchor regularization:

$$Q = \text{diag} \left(\frac{1}{U_m^2} + \lambda \right)_i, \quad (30)$$

where U_m is the utility of member m evaluated at $\mathbf{x}_{\text{conv}}$ (Eq. 3), the index i maps to member m via the variable layout, and λ is the fixed anchor weight. The term $1/U_m^2$ approximates the diagonal of the Nash welfare Hessian $\nabla^2 W$, and λ penalizes deviation from the decoder output $\boldsymbol{\theta}$.

Linear term $\mathbf{p}$.

$$\mathbf{p} = \lambda (\mathbf{x}_{\text{conv}} - \boldsymbol{\theta}), \quad (31)$$

where $\boldsymbol{\theta}$ is the decoder output. This term anchors the QP solution toward $\boldsymbol{\theta}$.

Nash correction weight. The relative influence of Nash coordination versus decoder fidelity is:

$$\gamma_{\text{Nash}} = \frac{1/U_m^2}{1/U_m^2 + \lambda}. \quad (32)$$

7.2 Inequality Constraints: $G \mathbf{d} \leq \mathbf{h}$

Inherited from Phase 1 (re-expressed in $\mathbf{d}$)

(C1') Variable bounds. Each variable must stay within its feasible range:

$$-d_i \leq x_i^{\text{conv}} - x_i^{\text{lb}}, \quad d_i \leq x_i^{\text{ub}} - x_i^{\text{conv}}. \quad (33)$$

This yields $2n$ inequality rows.

(C2') Time ordering. The temporal constraints $e_{m,t} \geq s_{m,t}$ and $s_{m,t+1} \geq e_{m,t}$ become:

$$d_{s_{m,t}} - d_{e_{m,t}} \leq e_{m,t}^{\text{conv}} - s_{m,t}^{\text{conv}}, \quad (34)$$

$$d_{e_{m,t}} - d_{s_{m,t+1}} \leq s_{m,t+1}^{\text{conv}} - e_{m,t}^{\text{conv}}. \quad (35)$$

These remain linear, as in Phase 1.

Family-level coordination constraints

(C4') Vehicle capacity. The vehicle capacity constraint $\sum_{m,t} d_{m,t} \cdot p_{m,t}^{\text{car}} \cdot \mathbb{1}_{\text{adult}}(m) \cdot a_{m,t} \leq n_{\text{veh}}$ involves the bilinear product $d_{m,t} \cdot p_{m,t}^{\text{car}}$ (both are decision variables), which cannot be directly included in a QP. It is linearized via first-order Taylor expansion around $\mathbf{x}_{\text{conv}}$:

$$\sum_{m,t} \left(p_{m,t}^{\text{car},0} \Delta d_{m,t} + d_{m,t}^0 \Delta p_{m,t}^{\text{car}} \right) \mathbb{1}_{\text{adult}}(m) a_{m,t} \leq n_{\text{veh}} - \sum_{m,t} d_{m,t}^0 p_{m,t}^{\text{car},0} \mathbb{1}_{\text{adult}}(m) a_{m,t}, \quad (36)$$

where superscript 0 denotes values at $\mathbf{x}_{\text{conv}}$, and Δ denotes the displacement component. This linearization is valid locally around $\mathbf{x}_{\text{conv}}$. It contributes a single linear inequality row. The same linearization structure is used in the SQP subproblem (Eq. 20), but evaluated at the current iterate $\mathbf{x}^{(i)}$ rather than $\mathbf{x}_{\text{conv}}$.

7.3 Equality Constraints: $\text{Ad} = \mathbf{b}$

Inherited from Phase 1 (re-expressed in $\mathbf{d}$)

(C3') Mode probability normalization. Since $\sum_k p_{m,t,k}^{\text{mode}} = 1$ holds at $\mathbf{x}_{\text{conv}}$, the displacement must preserve this:

$$\sum_{k=1}^K \Delta p_{m,t,k}^{\text{mode}} = 0, \quad \forall m, t. \quad (37)$$

Joint travel coordination

(C5') Joint travel consistency. When two members m_1, m_2 are both engaged in joint travel at activity t , their activity attributes should agree. The activation is determined by a sigmoid gate evaluated at $\mathbf{x}_{\text{conv}}$ and fixed as a binary decision:

$$w_{m_1, m_2, t} = \mathbb{1}[\sigma(10(j_{m_1, t}^0 \cdot j_{m_2, t}^0 - \tau)) > 0.1], \quad (38)$$

where $\tau = 0.5$ is the joint consistency threshold, $\sigma(\cdot)$ is the sigmoid function, and superscript 0 denotes values at $\mathbf{x}_{\text{conv}}$. The sigmoid of the bilinear product $j_{m_1, t} \cdot j_{m_2, t}$ is inherently nonlinear; evaluating it at $\mathbf{x}_{\text{conv}}$ and fixing the result as a binary gate converts this into a fixed set of linear equalities suitable for the QP. The same gate structure is used in the SQP subproblem (Eq. 21), but evaluated at the current iterate $\mathbf{x}^{(i)}$.

For each active pair ($w_{m_1, m_2, t} = 1$):

$$\Delta s_{m_1, t} - \Delta s_{m_2, t} = s_{m_2, t}^0 - s_{m_1, t}^0, \quad (39)$$

$$\Delta e_{m_1, t} - \Delta e_{m_2, t} = e_{m_2, t}^0 - e_{m_1, t}^0, \quad (40)$$

$$\Delta p_{m_1, t, k}^{\text{mode}} - \Delta p_{m_2, t, k}^{\text{mode}} = p_{m_2, t, k}^{\text{mode}, 0} - p_{m_1, t, k}^{\text{mode}, 0}, \quad \forall k, \quad (41)$$

$$\Delta j_{m_1, t} - \Delta j_{m_2, t} = j_{m_2, t}^0 - j_{m_1, t}^0. \quad (42)$$

Each active pair contributes $2 + K + 1 = 14$ equality rows. Note that the right-hand sides drive the paired variables toward agreement: if the pair is active and the values differ at $\mathbf{x}_{\text{conv}}$, the QP will force them to converge.

Table 2: Phase 2 constraint summary. All five constraints from Phase 1 are re-expressed in displacement space $\mathbf{d} = \mathbf{x} - \mathbf{x}_{\text{conv}}$.

Constraint	Phase 2 form	Space	Linearization method
(C1') Bounds	$G\mathbf{d} \leq \mathbf{h}$	$\mathbf{d}$	Direct translation
(C2') Time ordering	$G\mathbf{d} \leq \mathbf{h}$	$\mathbf{d}$	Direct translation
(C3') Mode simplex	$A\mathbf{d} = \mathbf{b}$	$\mathbf{d}$	Conservation (zero-sum)
(C4') Vehicle capacity	$G\mathbf{d} \leq \mathbf{h}$	$\mathbf{d}$	Taylor expansion at $\mathbf{x}_{\text{conv}}$
(C5') Joint consistency	$A\mathbf{d} = \mathbf{b}$	$\mathbf{d}$	Sigmoid gate fixed at $\mathbf{x}_{\text{conv}}$

7.4 Phase 1 $\rightarrow$ Phase 2 Connection

Table 2 summarizes the full constraint set in Phase 2.

7.5 QP Solve

Algorithm 2 Phase 2: QP Solve with Gradient Re-attachment

Require: $\mathbf{x}_{\text{conv}}$, $\boldsymbol{\theta}$ (decoder output), masks, utility parameters, V5: context-aware parameters, dynamic baseline

Ensure: $\mathbf{x}_{\text{final}}$ (with gradient to decoder, utility parameters, and UtilityParameterNet)

```

— Step 1: Construct QP parameters —
1:  $U_m \leftarrow \text{Eq. (3) at } \mathbf{x}_{\text{conv}}$  ▷ V5: uses ctx_params and dynamic baseline
2:  $Q_{\text{diag}} \leftarrow 1/U_m^2 + \lambda$  (expanded per variable via member mapping) ▷ Eq. (30)
3:  $\mathbf{p} \leftarrow \lambda(\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})$  ▷ Eq. (31)
— Step 2: Build constraints (C1'–C5) —
4:  $(G, \mathbf{h}) \leftarrow \text{Eqs. (33)–(36)}$  ▷ C1', C2', C4
5:  $(A, \mathbf{b}) \leftarrow \text{Eqs. (37)–(42)}$  ▷ C3', C5
— Step 2.5: Constraint preprocessing —
6: Remove all-zero rows from  $A$  and  $G$  ▷ Inactive constraints for masked variables
7:  $A \leftarrow \text{QR-RANK-FILTER}(A)$  ▷ Eq. (43); remove linearly dependent rows
8:  $\mathbf{h} \leftarrow \mathbf{h} + 10^{-4}$  ▷ Slight relaxation for numerical feasibility
— Step 3: Solve QP (under no_grad) —
9:  $\mathbf{d}^* \leftarrow \arg \min_{\mathbf{d}} \frac{1}{2} \mathbf{d}^\top Q \mathbf{d} + \mathbf{p}^\top \mathbf{d} \text{ s.t. } G\mathbf{d} \leq \mathbf{h}, A\mathbf{d} = \mathbf{b}$ 
10: if  $\mathbf{d}^*$  contains NaN or Inf then  $\mathbf{d}^* \leftarrow \mathbf{0}$ 
11: end if
— Step 4: Re-attach gradient —
12:  $\mathbf{d}^* \leftarrow \text{DIAGONALKKTBKWARD}(\mathbf{p}, Q_{\text{diag}}, \mathbf{d}^*)$  ▷ See §7.7
— Step 5: Assemble final output —
13:  $\mathbf{x}_{\text{final}} \leftarrow \boldsymbol{\theta} + (\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})_{\text{detach}} + \mathbf{d}^*$  ▷ Eq. (44)
14:  $\mathbf{x}_{\text{final}} \leftarrow \text{PROJECT}(\mathbf{x}_{\text{final}})$ 
15: if  $\mathbf{x}_{\text{final}}$  contains NaN then  $\mathbf{x}_{\text{final}} \leftarrow \boldsymbol{\theta}$  ▷ Full fallback
16: end if
17: return  $\mathbf{x}_{\text{final}}$ 

```

QR rank filtering. Before passing the equality constraint matrix A to the QP solver, linearly dependent rows are removed to prevent singular KKT systems. This is achieved via QR decomposition:

$$A^\top = QR, \quad \text{keep row } i \text{ if } |R_{ii}| > 10^{-8}. \quad (43)$$

The filtering uses the first batch element (constraint structure is similar across the batch) and is applied to both Phase 1 constrained SQP subproblems and the Phase 2 QP.

7.6 Final Solution Assembly

The final output is composed as:

$$\mathbf{x}_{\text{final}} = \boldsymbol{\theta} + \underbrace{(\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})}_{\text{no gradient}} + \underbrace{\mathbf{d}^*(\boldsymbol{\theta}, Q)}_{\text{carries gradient}}, \quad (44)$$

where the SQP displacement $(\mathbf{x}_{\text{conv}} - \boldsymbol{\theta})$ is treated as a constant (no gradient flows through Phase 1), and $\mathbf{d}^*$ depends on $\boldsymbol{\theta}$ through $\mathbf{p}$ (Eq. 31) and on utility parameters through Q (Eq. 30).

7.7 Diagonal KKT Backward

For the diagonal QP $\min_{\mathbf{d}} \frac{1}{2} \mathbf{d}^\top \text{diag}(Q_{\text{diag}}) \mathbf{d} + \mathbf{p}^\top \mathbf{d}$, the unconstrained closed-form solution is $\mathbf{d}^* = -\mathbf{p}/Q_{\text{diag}}$. We use this diagonal structure to approximate the gradients of the constrained solution:

$$\frac{\partial d_i^*}{\partial p_i} \approx -\frac{1}{Q_{\text{diag},i}}, \quad \frac{\partial d_i^*}{\partial Q_{\text{diag},i}} \approx -\frac{d_i^*}{Q_{\text{diag},i}}. \quad (45)$$

Given upstream gradient $\bar{\mathbf{d}}$ from the loss function, the gradients propagate as:

$$\bar{\mathbf{p}} = -\bar{\mathbf{d}} / Q_{\text{diag}} \quad \longrightarrow \text{gradient to } \boldsymbol{\theta} \text{ (decoder)}, \quad (46)$$

$$\bar{Q}_{\text{diag}} = -\bar{\mathbf{d}} \odot \mathbf{d}^* / Q_{\text{diag}} \quad \longrightarrow \text{gradient to utility parameters via } 1/U_m^2. \quad (47)$$

8 Straight-Through Log Unpack

After optimization, probability variables (p^{mode}, j, d) must be converted back to log-probabilities for the downstream cross-entropy loss. A naive $\log(p)$ introduces a $1/p$ factor in the gradient, causing instability for small probabilities.

The straight-through log estimator is applied:

$$\tilde{p}_k = p_k + (\log p_k - p_k)_{\text{detach}}, \quad (48)$$

so that the forward value equals $\log p_k$ while the backward gradient is the identity ($\partial \tilde{p}_k / \partial p_k = 1$). For binary heads (joint, driver), the scalar p is expanded to $[1 - p, p]$ before applying Eq. (48).

9 Learnable Parameters

Table 3 summarizes all parameters of the Nash Bargaining layer.

All learnable parameters are updated via backpropagation through the diagonal KKT backward (Eqs. 46–47). The UtilityParameterNet parameters ϕ_{MLP} receive gradients through the $Q_{\text{diag}} \rightarrow 1/U_m^2 \rightarrow U_m \rightarrow \alpha^{\text{ctx}}$ path (Eq. 11). The anchor weight λ is a fixed hyperparameter controlling the trade-off between Nash coordination and fidelity to the decoder output.

10 Two-Phase Flow Summary

1. **Pack:** Convert decoder predictions (logits) to probability-space vector $\boldsymbol{\theta}$.
2. **Phase 1 (SQP):** Starting from $\mathbf{x}_0 = \boldsymbol{\theta}$, iterate BFGS + Armijo to find $\mathbf{x}_{\text{conv}}$ (Alg. 1). Handles per-variable constraints (C1–C3) via projection.

Table 3: Parameters of the Nash Bargaining layer. Parameters marked with $\dagger$ can be replaced by per-activity context-aware predictions from the UtilityParameterNet (§3) when V5 is enabled.

Parameter	Dim	Initial Value	Description
β_0	1	2.0	Baseline utility (V5: replaced by dynamic baseline,
$\alpha_{\text{social}}^\dagger$	1	0.3	Out-of-home social preference
$\alpha_{\text{joint}}^\dagger$	1	0.1	Joint travel preference
$\theta_{\text{joint_cost}}$	1	-0.5	Joint coordination cost (quadratic, Eq. 4)
$\theta_{\text{escort}}^\dagger$	1	-0.58	Escort burden (adults)
θ_{travel}	$K=11$	$[-1.5, -0.4, \dots]$	Mode-specific travel time disutility
<i>V5: UtilityParameterNet (§3)</i>			
ϕ_{MLP}	$(F_f + F_m + 10) \times 64 + 64 \times 3$	$\approx$ global params	Context-aware parameter MLP
λ (fixed)	1	0.3	Anchor weight (not learned)

3. **Phase 2 (QP)**: Construct QP (Eq. 29) using $\mathbf{x}_{\text{conv}}$ and $\boldsymbol{\theta}$. Enforces all constraints: (C1'–C3') inherited from Phase 1, plus **new** family-level coordination constraints (C4) vehicle capacity and (C5) joint consistency, linearized at $\mathbf{x}_{\text{conv}}$. Gradient flows via diagonal KKT backward (Alg. 2).

4. **Project**: Enforce feasibility on $\mathbf{x}_{\text{final}}$.

5. **Unpack**: Convert back to prediction dict using straight-through log (Eq. 48).

Gradient flows from loss $\rightarrow$ unpack $\rightarrow \mathbf{x}_{\text{final}} \rightarrow \mathbf{d}^* \rightarrow$

- $\mathbf{p} \rightarrow \boldsymbol{\theta} \rightarrow$ decoder parameters,
- $Q_{\text{diag}} \rightarrow 1/U_m^2 \rightarrow U_m \rightarrow$ learnable utility parameters $(\alpha_{\text{joint}}, \alpha_{\text{social}}, \theta_{\text{escort}}, \theta_{\text{joint_cost}}, \beta_0, \theta_{\text{travel}})$.

V5 additional gradient paths. When context-aware parameters and dynamic baseline are enabled, two additional paths emerge:

- $Q_{\text{diag}} \rightarrow 1/U_m^2 \rightarrow U_m \rightarrow \alpha^{\text{ctx}} \rightarrow \phi_{\text{MLP}}$ (UtilityParameterNet parameters, Eq. 11),
- $Q_{\text{diag}} \rightarrow 1/U_m^2 \rightarrow U_m \rightarrow \theta_{\text{joint_cost}}$ (quadratic joint cost parameter).

Note that context-aware parameters are *detached* during Phase 1 (SQP); their gradients flow exclusively through Phase 2 (OptNet). The dynamic baseline $\beta_{0,m}$ is also detached (computed once at the decoder output) and does not receive gradients directly, but it influences the Q_{diag} values and thereby the optimization landscape.